NAME.....

SIGNATURE.....CLASS.....

545/1

CHEMISTRY.

PAPER 1.

2 hours

NEW LOWER SECONDARY EXAMINATION

END OF TERM ONE EXAMINATION 2026

PAPER 1

2 HOURS

INSTRUCTIONS:

- ★ *This paper consists of* **two** *sections* **A** *and* **B.** *It has* **six** *examinationitems.*

  Section **A** *has* **one compulsory** items.
- ★ *Section* **B** *has* **two parts, I** and **II.** *Answer* **one** item from each part. Answer **four** items in all.
- ★ *Answers to section* **A must** *be written in the spaces provided while those of section* **B must** *be written on the provided answer sheets.*
- ★ *Any additional item(s) answered will* **not** *bescored*

  Where necessary use , C = 12, H = , 1 mole of a gas occupies 22.4 litres at s.t.p

SECTION A

Answer **all** *items in this section in the spaces provided.*

ITEM 1

When Hamza felt sick, his mother prepared an aloe vera extract for him which provided a brief improvement on first day. However, his symptoms persisted the next day. He then went to a nearby facility where he was diagnosed with *gastrointestinalpoisoning* known as *salmonellainfection* resulting from taking contaminated food. He was given a product which he used and recovered.

Task. As a chemistry student,

(a) State the type of the product used which enabled him to completely recover and how it works.

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..... (b) advise him about any danger associated with the product he used.

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(c) evaluate the substances he used for treatment from the time he felt sick to the time of recovery.

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ITEM 2

A production industry wants to convert ethene, an excess waste gas, from its production processes into another useful product, ethane. They plan to react the hydrocarbon with hydrogen under suitable controlled conditions. Below is the equation of reaction.

$$C_2H_4(g) + H_2(g) \longrightarrow C_2H_6(g)$$

The manager needs to know the mass of product in Kg that can be produced when 10,000 litres of ethene at s.t.p reacts with hydrogen . He also needs to know the environmental concern associated with the product. You have been invited for guidance

Task. As a learner of chemistry,

(a) explain the category of each hydrocarbon in the reaction.

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(b) Suggest the suitability of product to its applications in every day

life. .....

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..... (c) Calculate the mass of product formed in above reaction .

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(d) explain the environmental impact associated with the product.

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SECTION B

Part I (Attempt *one item from this part)*

ITEM 4

The demand for aluminium by several industries dealing in the production of kitchenware and duralumin alloy has increased. This has prompted the government to clear a local investor to establish an aluminium extraction plant in one district rich in aluminium reserves. However, the locals would like to know how the extraction will be carried out. They have also raised concerns about how they will benefit from this investment and environmental responsibility. You have been invited to talk to the locals in one of local radio stations organized by district officials .

Task. As a chemistry student, make a write-up you will use during the presentation.

ITEM 4.

The government through Uganda Investments Authority has cleared an investor to set up a cement production facility in one district of Uganda. However, the locals are concerned about its impact resulting from the process of production. You have been invited to address the locals in one of the sensitization workshops conducted at the sub county hall

Task. Write a write up presentation you will use during the presentation.

Part II.

(***Attempt one item from this part.***)

ITEM 5.

Many water sources and forested regions in Uganda have been negatively impacted due to urban development and increased population. This has prompted many environmental bodies to raise awareness campaigns among the public on the need for environmental protection and sustainable use. You have been invited to one of the sensitization workshops.

Task. As a chemistry prepare a short presentation you will deliver in the radio talk show.

ITEM 6

Upon the discovery of the coal reserves in the some parts of kigezi region, the locals raised environmental impact associated with its exploitation. They also lacked comprehensive knowledge of importance of coal . This has prompted the government to conduct a regional workshops to raise awareness amongst the public to be informed about crude oil . You have been invited to attend address the public.

Task. As a chemistry student, prepare a short message you will use to deliver in the workshop.

END

YOU MAY DELAY BUT TIME WAITS FOR NO MAN